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current for the inbuilt driver transistor of IC1. This current should be below 100mA. Operating frequency of MC34063A is in the range of 40kHz to 60kHz, and depends mainly on capacitor C3. Inductor L1 is in the range of 100μH to 150μH for current above 1.8A.

Output voltage is around 48V and can be adjusted with R4. It can be calculated using the approximate relationship:

$$V_{out} = 1.25V \times (1 + R4/R5)$$

Output current is up to around 50mA, which is enough for several applications including microphone amplifiers. Voltage on pin 5 of IC1 should be equal to internal reference voltage or around 1.25V (1.21V to 1.29V). Schottky diode D2 should preferably be rated at least 100V. Schottky or ultrafast diode can either be SB1A0, SR1100, SS110 or SR180.

C4 and C5 are the first filtering capacitors. Ripples at OUTPUT1 are around 50mV at 50mA load, which may be too high for some applica-

PARTS LIST

Semiconductors:

IC1	- MC34063A switching regulator
D1, D3	- 1N4007 rectifier diode
D2	- SR1100 Schottky diode
LED1, LED2	- 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

R1	- 1.5-kilo-ohm
R2	- 0.22-ohm
R3	- 68-ohm
R4	- 39-kilo-ohm
R5	- 1-kilo-ohm
R6	- 22-kilo-ohm

Capacitors:

C1	- 220μF, 16V electrolytic
C2, C4	- 100nF ceramic disk
C3	- 680pF ceramic disk
C5, C6	- 220μF, 63V electrolytic

Miscellaneous:

CON1-CON3	- 2-pin terminal connector
F1	- 1A fuse with holder
L1	- 100μH inductor
L2	- 33μH inductor
	- 5V power supply

tions. So, second output (OUTPUT2) is provided after LC filter. L2 (usually from 22μH to 220μH for current above

100mA) and C6 produce additional filtration of the voltage produced by the DC-to-DC converter.

We can use both the outputs, OUTPUT1 and OUTPUT2, at the same time under the condition that total current is below 60mA. The circuit does not need any adjustment.

Construction and testing

An actual-size PCB layout of the DC-to-DC converter is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB, connect 5V regulated power supply across CON1. Connect LED1 and LED2 on the front panel to indicate power status. **EFY**



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